

Deliberation and Learning: Evidence From Deliberative Polls

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Working draft

Abstract

Deliberative Polls are meant to produce more informed opinion. We ask how much participants learn, whether taking part causes the learning, and who learns. In four Deliberative Polls with control groups, held in seven countries, participation raised factual knowledge by .46 standard deviations (95% interval .34 to .58); in the one poll followed for a year, 45% of the effect remained. Across 22 polls without controls, the share of items answered correctly rose by .130 [.099, .162], about twice as much face to face as online. A widely used correction for lucky guesses finds nearly as much learning among untreated controls as among participants; a latent class model does not. Participants who start out knowing less, and those without degrees, gain somewhat more from participation. Small groups are composed as if at random, and who is in one's group, including how much the others know, has no detectable effect on learning.

1 Introduction

Deliberative Polls teach. In four polls that surveyed a control group alongside the participants—two in the United States, one in Tanzania, and one across six middle-income countries—taking part raised factual knowledge by .46 standard deviations of the controls' initial knowledge. In the one poll re-interviewed a year later, 45% of that effect remained.

Learning is among the great virtues commonly ascribed to deliberation. Participants learn relevant facts, the arguments for and against the alternatives, and what people unlike themselves think. This helps them discern which policies accord with their values and interests, and it helps majorities know what they are doing. Past analyses of Deliberative Polls found that participants learn a lot (Luskin et al. 2002). But they compared participants before and after, without a comparison group, so they could not say how much of the learning deliberation caused, or who learns.

We answer both questions and correct two measurement practices along the way. First, the effect of participation is large, positive and durable in every study with a control group. Second, the standard correction for lucky guesses on knowledge items manufactures learning: applied to control groups that received no treatment, it finds gains of .11 to .13 of the scale, nearly the size of the true effects. The latent class model of Cor and Sood (2016) passes the same test. Third, models of post-deliberation knowledge that use the corrected pre-deliberation score build the outcome into a regressor; with pre-deliberation measures, the estimated weight of initial knowledge falls by nearly half. Fourth, small-group assignment is as good as random, so group composition is a natural experiment, and it shows no effect of one's groupmates on how much one learns.

2 Deliberative Polling and Learning

Deliberative Polling (Fishkin 1995; Luskin et al. 2002) aims to capture fuller-information, more thoughtful opinions than ordinary polls record. A Deliberative Poll interviews a random sample, assembles as many as are willing to deliberate a policy or electoral choice in alternating small-group discussions and plenary sessions with panels of competing experts, and then re-measures the sample’s opinions.

The design affords unusual opportunities and incentives to learn. Participants receive balanced briefing documents that rehearse the arguments each side would make and embed each side’s pet facts. They know from recruitment that they will attend an Event, capital “e”—away from home, often televised, often attended by prominent policy experts or policy-makers—and many use the intervening weeks to prepare, reading the briefing materials, following the topic in the news, and talking about it with friends, family, and coworkers. If knowledge depends on the confluence of cognitive ability, motivation, and exposure to relevant information (Luskin 1990), Deliberative Polling stands to increase two of the three.

3 Data

Table 1 lists the data. For 22 polls held between 1994 and 2009 we have each participant’s proportion of knowledge items answered correctly before (T1) and after (T2) deliberating, with small-group membership and demographics: 21 from the replication data for Luskin et al. (2022), and a poll in Marousi, Greece, in which a party chose its mayoral candidate, released here for the first time; for 23 polls, most of them the same, we have responses item by item (Cor and Sood 2016). Four more recent polls surveyed control groups.

Table 1: Data. Participants and controls are panel respondents used in the analyses. Control groups are an uninvited random sample, or randomized; in the antimicrobial-resistance poll the treated group is the attendees.

Study	Design	Participants	Controls
21 Deliberative Polls, 1994–2009	Pre-post	5,867	
Marousi, Greece, 2006	Pre-post	146	
23 Deliberative Polls, item level	Pre-post	6,654	
America in One Room, 2019	Uninvited sample	523	844
America in One Room: Climate, 2021	Uninvited sample	962	671
Tanzania, 2015	Randomized	383	921
Antimicrobial Resistance, 2024	Randomized; attendees	1,280	1,139

The control groups differ in how they were formed. America in One Room and its 2021 climate successor interviewed a separate random sample that was never invited (Fishkin et al. 2021). The Tanzanian poll randomized villages to treatment or control and, within treated villages, individuals to deliberation or to the briefing materials alone (Sandefur et al. 2022). The poll on antimicrobial resistance, held online in Brazil, Colombia, India, Indonesia, Nigeria and Tanzania, randomized panel members after the first interview; the release contains the controls and the invitees who attended, but not the invitees who stayed home (Mendelson et al. 2026).

Appendix Table 3 lists our recodes. Three matter for earlier results: the National Issues Convention

was coded as online although it met in Austin, its age variable holds birth years, and the knowledge-based group variable is corrupted for Australia.

4 Measuring Learning

We score each closed-ended knowledge item 1 if answered correctly and 0 if answered incorrectly or with “don’t know” (Luskin and Bullock 2011), and average across items. A participant’s observed gain is $K_2 - K_1$, where K is observed knowledge, subscripted for wave.

Observed gain is a poor proxy for actual gain at the individual level (Luskin et al. 2016). Participants who know a lot at the outset cannot show much increase, because they are near the top of the scale, yet they are often those actually learning the most. K_2 , by contrast, is high for everyone learning a lot, whether they started low and observably learned or started high and unobservably learned. So when explaining who learns we model K_2 with K_1 among the regressors, Allison’s (1990) “regressor variable” approach, and when describing how much a poll’s participants learned we report the mean gain.

Closed-ended items invite guessing, which inflates K_1 and can hide learning. Two corrections are common. The first, used in our earlier drafts, assumes that a participant who answers an item correctly at T1 and incorrectly at T2 guessed luckily at T1, and resets that T1 answer to 0. The second fits a latent class model to each item’s pattern of answers across the two waves, separating respondents who knew, learned, or guessed (Cor and Sood 2016). Both assume nobody forgets.

5 How Much Do Participants Learn, and Does Participating Cause It?

Figure 1 puts all the evidence on one scale, standard deviations of T1 knowledge. Panel A shows the mean gain in each of the 22 polls without a control group; panel B shows the effect of participation in the four polls with one.

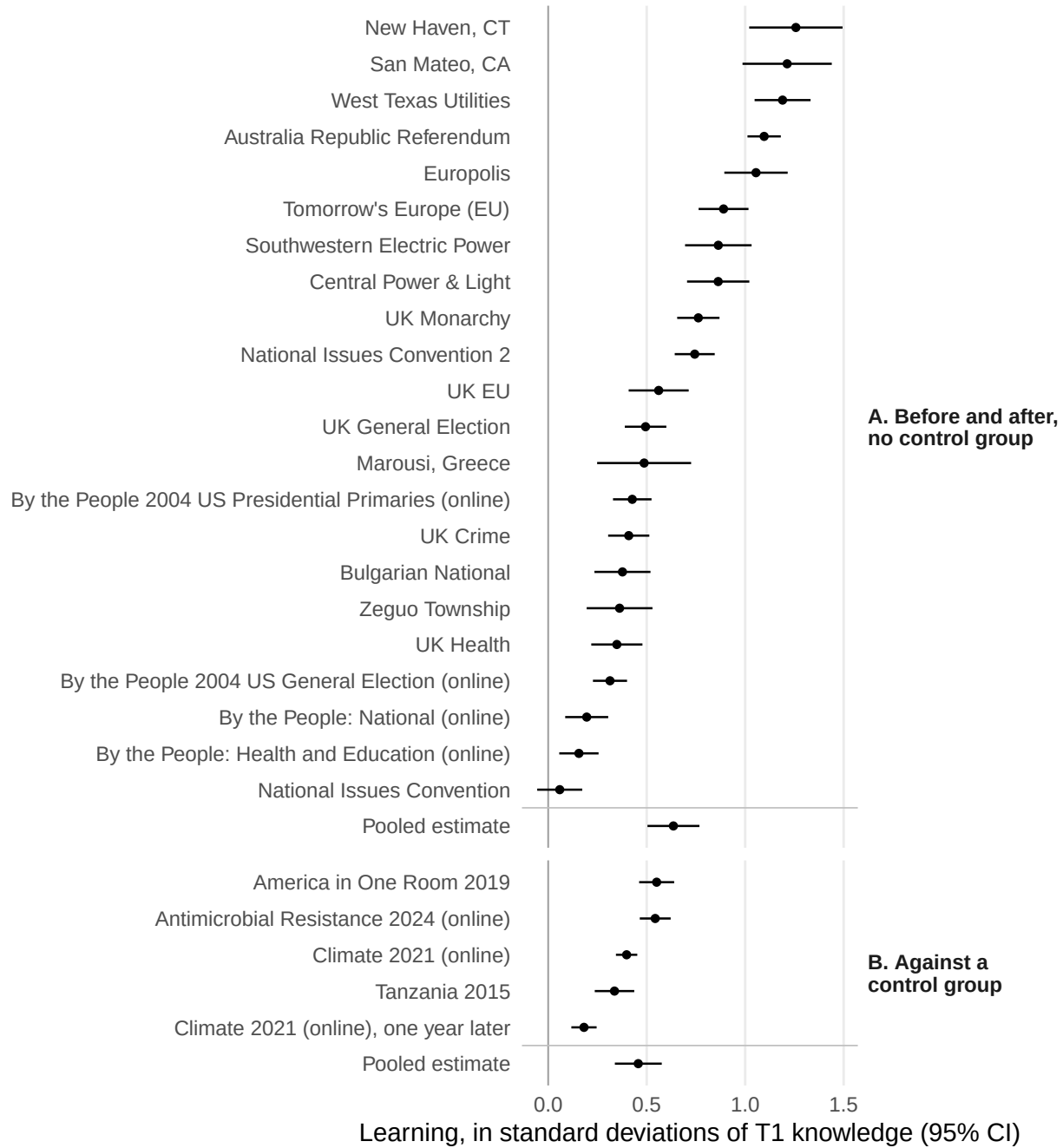


Figure 1: Learning in every poll, in standard deviations of T1 knowledge, with 95% intervals. A: mean T2 minus T1 gain among participants, scaled by the poll's T1 standard deviation. B: effect of participation, from a regression of T2 on treatment and T1 knowledge, scaled by the control group's T1 standard deviation; Tanzanian standard errors are clustered by village. Pooled estimates are from Bayesian random-effects meta-analyses.

Every poll shows learning. Raw gains range from .012 to .288 of the items; pooled, the proportion correct rose by .130 [.099, .162], or .64 [.50, .77] standard deviations. The pooling is a Bayesian random-effects meta-analysis: each poll's mean gain is its true gain plus sampling error, true gains are normally distributed around an overall mean, the prior on the mean is flat and that on the

between-poll standard deviation half-normal with scale .1 (Röver 2020). Face-to-face polls produced about twice the gain of online ones (.145 [.113, .177] against .065 [-.001, .132]; Appendix Table 5).

A pre-post gain alone does not show that deliberation taught anything: participants take the same quiz twice and live through weeks of news. The control groups net these out, and participation still teaches. Adjusting for T1 knowledge, it added .142 (.012) to the proportion correct in America in One Room, .121 (.008) in the 2021 climate poll, and .125 (.009) in the antimicrobial-resistance poll, where the country estimates range from -.04 in Tanzania to .31 in Nigeria. In Tanzania’s national poll, deliberation raised a standardized knowledge index by .335 (.051) standard deviations, matching the published estimate (Sandefur et al. 2022). Pooled, the effect is .46 [.34, .58] standard deviations. A year after the climate poll, attendees still knew .055 more than controls (s.e. .010), 45% of the immediate effect. The controls’ own gains, .01 to .03 of the items, show how little of a pre-post gain testing and news explain.

The designs differ in what they rule out. Tanzania randomized; its estimates are causal. The antimicrobial-resistance poll randomized too, but its treated group is the attendees, who were 23 to 92% of invitees depending on the country, so the comparison adds self-selection to randomization. The two American polls compare attendees with a separately drawn sample, and attendees are a selected subset of invitees: in America in One Room they answered .458 of T1 items correctly against .378 for invitees who stayed home, and in the climate poll .686 against .579. Conditioning on T1 knowledge removes differences in where people start. The difference in gains, which instead assumes that the two groups would have gained equally, gives .131, .093 and .103 in the three studies; when both assumptions are plausible the truth lies between the two estimates (Angrist and Pischke 2009).

Does deliberation add anything to the briefing materials? In Tanzania, deliberation and the briefing materials alone were randomized among individuals in the same villages. Deliberation added .205 (.058) standard deviations adjusting for T1 knowledge, but only .081 (.081) in the difference of gains, because the deliberation arm happened to start .16 standard deviations higher. Discussion probably helps; this one comparison cannot say by how much.

6 What the Guessing Correction Gets Wrong

Control groups offer a test of the corrections: nobody in them was treated, so a correction that finds learning there is finding an artefact. Figure 2 applies both corrections to attendees and controls in the three studies with item-level data.

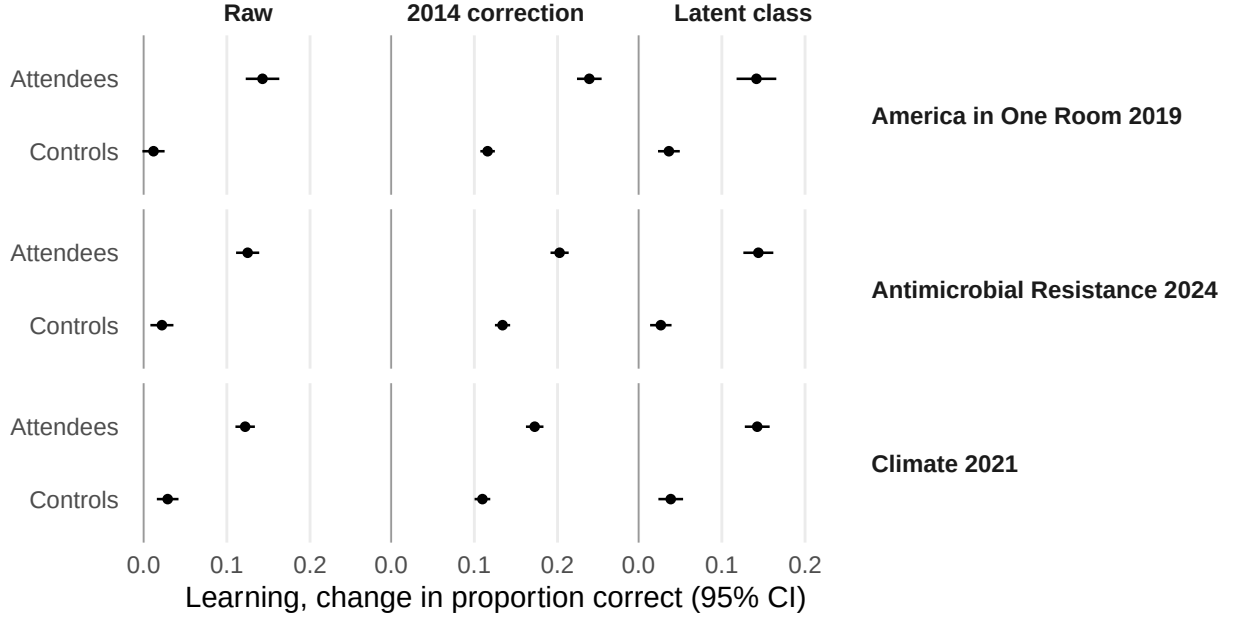


Figure 2: Learning estimated three ways among attendees and untreated controls, with 95% intervals. Raw: T2 minus T1 proportion correct. 2014 correction: T1 answers that are correct but wrong at T2 are reset to incorrect. Latent class: mean item learning from the model of Cor and Sood (2016). Intervals for raw and corrected gains use person-level standard errors; latent class intervals use 200 bootstrap resamples of respondents.

The reset rule fails. Among controls it finds gains of .110 to .134, against raw gains near zero, because 14 to 37% of the controls' correct T1 answers are wrong at T2. Those reversals reflect inconsistent answering and forgetting, not lucky guesses later exposed. The latent class model, which estimates guessing from the whole pattern of answers, finds .027 to .039 among controls, and its attendee-minus-control differences match the raw ones. Forgetting is real: a year after the climate poll, attendees answered .08 fewer of the items correctly than at its end.

So the latent class estimates are the better guide to how much participants know after a poll: across the 23 polls with item-level data they average .208, against a raw .159.

7 Who Learns?

7.1 Whom participation helps most

The control-group studies let us ask whose knowledge participation raises most, a causal question the pre-post data cannot answer. Adding the product of treatment and centred T1 knowledge, the effect of participation is smaller for those who start knowing more, by -.15 (.04) per unit of T1 knowledge in America in One Room and -.25 (.03) in the climate poll, with no difference in the other two studies (Appendix Table 6). Part of this is the ceiling: those who start high have less room to show a gain. Participation also helped those without a bachelor's degree somewhat more in the climate and antimicrobial-resistance polls (-.047 (.016) and -.043 (.020)), though not in America in One Room. Deliberation narrows knowledge gaps modestly; it does not widen them.

7.2 Who ends up knowing the most

Within the 22 polls we model post-deliberation knowledge as a function of pre-deliberation knowledge, education, demographics, the small group, and the poll, with random intercepts for small groups and polls (Raudenbush and Bryk 2002; Bates et al. 2015). The main model uses 5,827 participants in 412 groups in 22 polls (Table 2).

Every regressor is measured before deliberation. Our earlier drafts used the guessing-corrected T1 score, which resets items answered wrong at T2 and so builds part of the outcome into the regressor; they also measured group knowledge with corrected scores, which contain the other members' T2 answers. With those measures the coefficient on T1 knowledge is .872; with pre-deliberation measures it is .482 (Appendix Table 4). Measurement error in raw T1 scores pulls the second estimate toward zero, so the true weight lies between the two, but the first is not an estimate of it.

Table 2: Post-deliberation knowledge: maximum likelihood estimates (standard errors) with random intercepts for small group and poll. The second column is limited to the 8 polls whose item-level answers link to small groups, which allows group knowledge to be measured on the items each participant missed at T1. The third is limited to the 4 polls that asked about the briefing materials and uses poll fixed effects in place of poll-level variables.

	All polls	8 polls, item-specific group knowledge	4 polls, b
Intercept	.053 (.049)	.070 (.069)	.519
T1 knowledge	.482 (.025)	.583 (.047)	.515
High school	.122 (.016)	.210 (.027)	.078
BA or more	.195 (.018)	.275 (.029)	.158
T1 knowledge x high school	-.052 (.029)	-.218 (.052)	-.088
T1 knowledge x BA or more	-.102 (.032)	-.266 (.055)	-.125
Age (decades)	-.007 (.002)	-.009 (.003)	-.014
Attitude extremity	-.007 (.018)	.019 (.020)	-.121
Group size	.000 (.001)	.001 (.003)	-.005
Group knowledge: others' mean T1	.086 (.051)		-.205
Group knowledge: others' T1 on items missed		-.058 (.044)	
Opinion heterogeneity	.024 (.082)	.002 (.123)	.118
Female	-.025 (.023)	-.093 (.040)	-.049
Proportion female in group	-.011 (.034)	-.071 (.056)	-.047
Female x proportion female	.030 (.044)	.140 (.072)	.069
Online	-.099 (.033)	-.148 (.047)	
Poll mean T1 knowledge	.517 (.104)	.568 (.119)	
Read briefing materials			.065
Participants	5,827	2,108	
Small groups	412	150	
Polls	22	8	

Education and initial knowledge predict who ends up knowing the most, and they substitute for each other: a bachelor's degree adds .195 (.018) for a participant who starts with nothing right, and less for those who start knowing more (-.102 (.032) per unit of T1 knowledge). Participants in online polls end up knowing less (-.099 (.033)), and those in polls whose participants start out

knowing more end up knowing more (.517 (.104)). In the four polls that asked, those who read more of the briefing materials knew more afterwards (.065 (.016)), though reading is itself a choice.

7.3 Do small groups matter?

If participants learn from each other, those placed with more knowledgeable groupmates should learn more. Estimating this requires that groupmates' characteristics be unrelated to one's own, which random assignment to groups ensures and self-selection does not. Organizers intend random assignment. We test it by regressing each participant's characteristic on the mean among the other members of the group, controlling for the mean among the other members of the poll, without which the test is biased toward finding negative sorting (Guryan et al. 2009). The coefficients are near zero for T1 knowledge (.017 (.017)), gender (-.016 (.020)) and age (-.008 (.025)), and small for a bachelor's degree (.042 (.030)). Groups are composed as if at random.

That makes group composition a natural experiment. Within each of 23 polls, the 22 above and America in One Room, we regress T2 knowledge on the other members' mean T1 knowledge, controlling for one's own and the poll's, and pool the estimates (Figure 3). The pooled effect is .069 [-.047, .190]: at the upper end, a group whose other members answered ten more points of the T1 items correctly would raise one's T2 score by 1.9 points. The share of women in the group has no detectable effect either (.027 [-.013, .066]). Measured on the items each participant missed, which is closer to what one could learn from others, group knowledge again has no effect (-.058 (.044) in the 8 polls with linked item data). Our earlier drafts reported a positive effect of that measure; it came from building it with T2 answers, which reflect the group's common experience (Manski 1993).

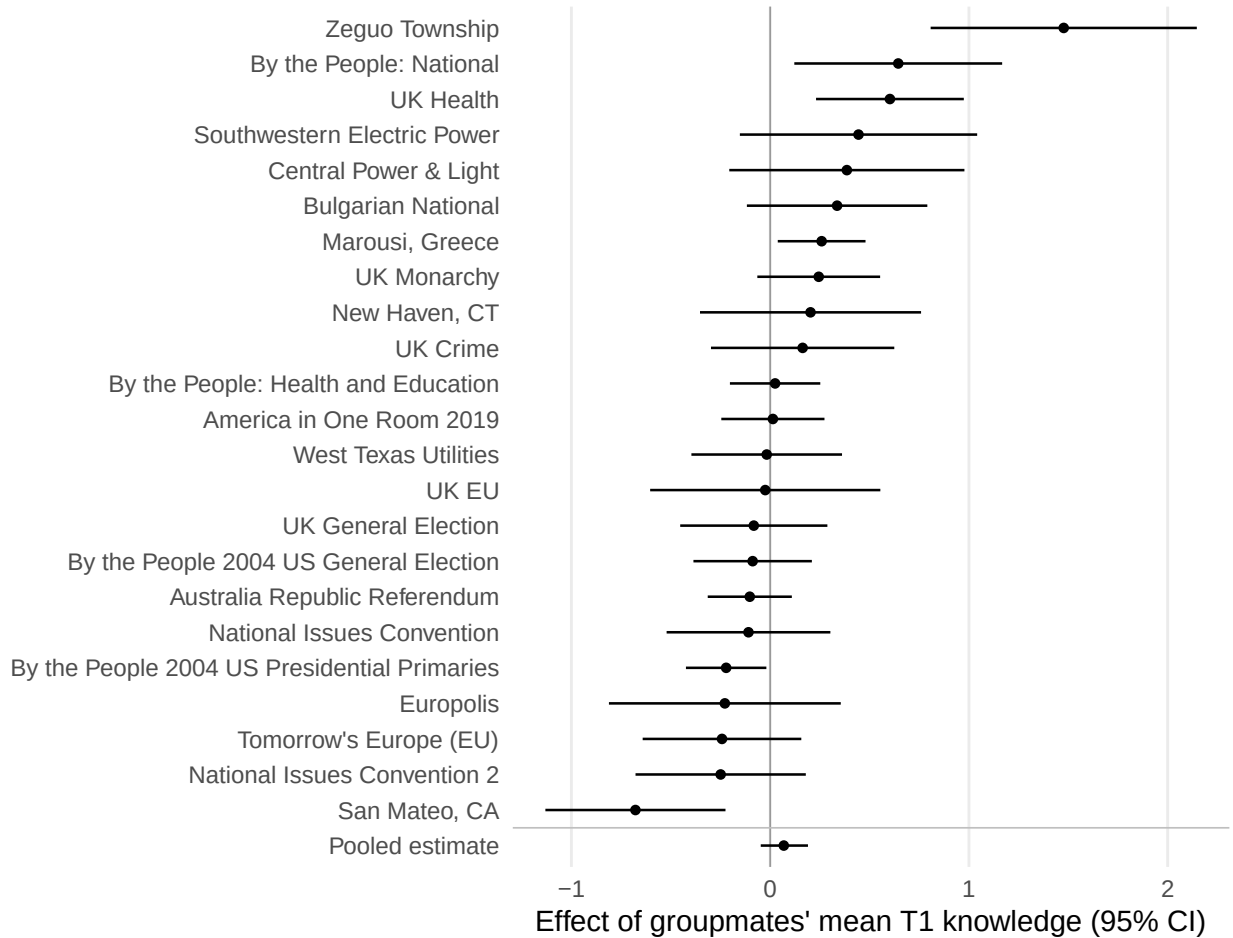


Figure 3: Effect of the other group members' mean T1 knowledge on own T2 knowledge, by poll, with 95% intervals. Each estimate controls for own T1 knowledge and the poll's leave-one-out mean, with standard errors clustered by small group; the pooled estimate is from a Bayesian random-effects meta-analysis.

8 Conclusion

Deliberative Polls raise what participants know by about half a standard deviation, and much of the gain lasts. The learning does not depend on whom participants happen to deliberate with, and it favours, if anything, those who start out knowing less. The simplest reading is that the whole event teaches, through the briefing materials, the expert sessions, the discussion, and the anticipation of taking part, rather than through particular groupmates.

Two limits remain. Only Tanzania randomized people to take part or not; elsewhere the comparison involves attendees, who differ from those who stay home. And the knowledge items are few and chosen by the organizers, so the gains are gains in the facts the event emphasized.

References

Allison, Paul D. 1990. "Change Scores as Dependent Variables in Regression Analysis." *Sociological Methodology* 20: 93. <https://doi.org/10.2307/271083>.

- Angrist, Joshua D., and Jörn-Steffen Pischke. 2009. *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press.
- Bates, Douglas, Martin Mächler, Ben Bolker, and Steve Walker. 2015. "Fitting Linear Mixed-Effects Models Using lme4." *Journal of Statistical Software* 67 (1). <https://doi.org/10.18637/jss.v067.i01>.
- Cor, M. Ken, and Gaurav Sood. 2016. "Guessing and Forgetting: A Latent Class Model for Measuring Learning." *Political Analysis* 24 (2): 226–42. <https://doi.org/10.1093/pan/mpw010>.
- Fishkin, James S. 1995. *The Voice of the People: Public Opinion and Democracy*. Yale University Press.
- Fishkin, James, Alice Siu, Larry Diamond, and Norman Bradburn. 2021. "Is Deliberation an Antidote to Extreme Partisan Polarization? Reflections on 'America in One Room'." *American Political Science Review* 115 (4): 1464–81. <https://doi.org/10.1017/s0003055421000642>.
- Guryan, Jonathan, Kory Kroft, and Matthew J Notowidigdo. 2009. "Peer Effects in the Workplace: Evidence from Random Groupings in Professional Golf Tournaments." *American Economic Journal: Applied Economics* 1 (4): 34–68. <https://doi.org/10.1257/app.1.4.34>.
- Luskin, Robert C. 1990. "Explaining Political Sophistication." *Political Behavior* 12 (4): 331–61. <https://doi.org/10.1007/bf00992793>.
- Luskin, Robert C., and John G. Bullock. 2011. "'Don't Know' Means 'Don't Know': DK Responses and the Public's Level of Political Knowledge." *The Journal of Politics* 73 (2): 547–57. <https://doi.org/10.1017/s0022381611000132>.
- Luskin, Robert C., James S. Fishkin, and Roger Jowell. 2002. "Considered Opinions: Deliberative Polling in Britain." *British Journal of Political Science* 32 (3): 455–87. <https://doi.org/10.1017/s0007123402000194>.
- Luskin, Robert C., Ariel Helfer, and Gaurav Sood. 2016. "Measuring Learning in Informative Processes." https://www.gsood.com/research/papers/k_gains.pdf.
- Luskin, Robert C., Gaurav Sood, James S. Fishkin, and Kyu S. Hahn. 2022. "Deliberative Distortions? Homogenization, Polarization, and Domination in Small Group Discussions." *British Journal of Political Science* 52 (3): 1205–25. <https://doi.org/10.1017/s0007123421000168>.
- Manski, Charles F. 1993. "Identification of Endogenous Social Effects: The Reflection Problem." *The Review of Economic Studies* 60 (3): 531. <https://doi.org/10.2307/2298123>.
- Mendelson, Marc, Alice Siu, Louise Gough, Hamish Morrow, James Fishkin, and Sally Davies. 2026. "Understanding What Citizens Think about Antimicrobial Resistance: Deliberative Polling® in Six Middle-Income Countries." *Wellcome Open Research* 10 (May): 690. <https://doi.org/10.12688/wellcomeopenres.24803.2>.
- Raudenbush, Stephen W., and Anthony S. Bryk. 2002. *Hierarchical Linear Models: Applications*

and Data Analysis Methods. 2nd ed. Sage.

Röver, Christian. 2020. “Bayesian Random-Effects Meta-Analysis Using the Bayesmeta r Package.” *Journal of Statistical Software* 93 (6). <https://doi.org/10.18637/jss.v093.i06>.

Sandefur, Justin, Nancy Birdsall, James Fishkin, and Mujibu Moyo. 2022. “Democratic Deliberation and the Resource Curse: A Nationwide Experiment in Tanzania.” *World Politics* 74 (4): 564–609. <https://doi.org/10.1017/s0043887122000090>.

A Recodes

Table 3: Recodes applied to the participant-level data

Poll	Variable	Rule	Rows affected	Evidence
National Issues Convention	mode	online 1 -> 0	all rows	Held face-to-face in Austin, January 1996 (Cor-Sood dpdata.csv; Luskin, Fishkin and Jowell 1998)
National Issues Convention	ppage	age = 1996 - ppage	all rows	ppage ranges 1900-1989 in this poll only, i.e. birth year
All polls	ppage	age < 16 or > 100 -> NA	Zeguo age 1; Euro- polis age 109	Deliberative Polls sample adults
BTP 2004 Presidential Primaries	all	exact duplicate rows dropped	217 of 434 rows	read_polardata() distinct(); identical to pk/data duplication
Australia Republic Referendum	grpgain	all rows -> NA	347 rows	49 values above 1 (max 5.82) and one Inf on a 0-1 scale; the 2014 estimate is recovered only without them (evidence/parity_2014.csv)
Marousi, Greece	t2know	0 -> NA when T1 > 0	26 of 146 rows	27 T2 scores of 0 against 2 at T1; dropping them reproduces the 2011 meta-analysis draft (n = 129, gain .08)
Marousi, Greece	educ3	educ4 code 7 -> NA	20 rows	Code outside the 0-1 scale; agg_data had mapped it to high school
Marousi, Greece	t1knowcor	all rows -> NA	146 rows	Values fall below zero (minimum -0.10) when raw T1 is 0, so this is not the reset correction used for the other polls
All polls	t1know, t2know, t1knowcor	rounded to 10 decimals	2 NIC rows	Stored as 1.0000000002; removes floating-point excess over 1

B Reproducing the 2014 estimates

We refit the 2014 draft’s main model on the file it used. That file was re-saved after the draft’s tables were produced: one poll’s rows were duplicated, and 50 Australian values of the group-knowledge variable were corrupted, reaching 5.82 on a 0–1 scale. Dropping both, the refit uses 3,811 participants against the draft’s 3,861, and all 19 coefficients agree with the published table

within .024. Rebuilding the group-knowledge variable from item-level data reproduces the archived values.

Table 4: The main model and the 2014 specification, which uses guessing-corrected T1 knowledge, the 2014 group-knowledge measure, and minority status.

	Pre-deliberation measures	2014 specification
Intercept	.053 (.049)	.251 (.044)
T1 knowledge	.482 (.025)	.872 (.022)
High school	.122 (.016)	.082 (.011)
BA or more	.195 (.018)	.130 (.013)
T1 knowledge x high school	-.052 (.029)	-.110 (.026)
T1 knowledge x BA or more	-.102 (.032)	-.189 (.028)
Age (decades)	-.007 (.002)	-.010 (.002)
Attitude extremity	-.007 (.018)	-.006 (.015)
Group size	.000 (.001)	.000 (.001)
Group knowledge: others' mean T1	.086 (.051)	
Group knowledge: 2014 measure		.042 (.026)
Opinion heterogeneity	.024 (.082)	-.002 (.069)
Female	-.025 (.023)	-.034 (.021)
Proportion female in group	-.011 (.034)	.006 (.029)
Female x proportion female	.030 (.044)	.072 (.039)
Minority		-.011 (.013)
Proportion minority in group		-.031 (.028)
Minority x proportion minority		-.059 (.041)
Online	-.099 (.033)	-.073 (.030)
Poll mean T1 knowledge	.517 (.104)	.130 (.098)
Participants	5,827	4,402
Small groups	412	314
Polls	22	17

C Further results

Table 5: Bayesian random-effects meta-analysis of poll mean raw gains

Model	Parameter	Median	2.5%	97.5%
Pooled	Mean gain	.130	.099	.162
Pooled	Between-poll SD	.070	.050	.096
By mode	Face-to-face	.145	.113	.177
By mode	Online	.065	-.001	.132
By mode	Between-poll SD	.064	.045	.088
raw SD, pooled	Mean gain	.636	.504	.768
raw SD, pooled	Between-poll SD	.300	.234	.375
raw SD, by mode	Face-to-face	.718	.585	.852
raw SD, by mode	Online	.273	-.004	.551
raw SD, by mode	Between-poll SD	.271	.207	.346

Table 6: Heterogeneous effects of participation. Each row adds the product of treatment and the moderator to the regression of T2 on treatment and T1 knowledge.

Study	Moderator	Treatment x moderator	n
America in One Room 2019	T1 knowledge (centred)	-.155 (.041)	1,367
America in One Room 2019	Bachelor's degree	.027 (.023)	1,367
America in One Room: Climate 2021	T1 knowledge (centred)	-.252 (.029)	1,633
America in One Room: Climate 2021	Bachelor's degree	-.047 (.016)	1,633
Antimicrobial Resistance 2024	T1 knowledge (centred)	.043 (.038)	2,419
Antimicrobial Resistance 2024	Bachelor's degree	-.043 (.020)	2,408
Tanzania 2015	T1 knowledge (centred)	-.018 (.051)	1,304

D Knowledge items

The items for the polls analyzed in the 2014 draft, with correct answers in parentheses where the item is not multiple choice.

D.1 UK (EU)

(All questions are true/false)

- EU has recently expanded to 15 members
- Switzerland is to join EU (false)
- Britain's income tax rates are decided in Brussels (false)
- Elections to European Parliament are held every 5 years
- Of the three British parties, Liberal Democrats least in favour of EU.

D.2 UK (Health)

(All questions are true/false)

- Standard charge for NHS prescription is 10 pounds (false)
- Proportion of old people in Britain getting larger (true)
- Even with inflation taken into account, spending on NHS doubled over last 20 years (true)
- Most people use private medical treatment instead of NHS (false)
- All British women get free breast cancer screening on NHS (false)
- Cosmetic surgery is never available on NHS (false)

D.3 UK (GE)

(First three questions are true/false. Others are party placement on a 7 point scale.)

- Inflation has been less than 5% over the past few years (true).
- Interest rates are decided by Bank of England (false).
- Unemployment in UK is higher than in Germany (false).

(Conservative, Labor, Liberal Democrat) Party's View: Government should do nothing to make incomes in Britain more equal.

(Conservative, Labor, Liberal Democrat) Party's View: Government should spend much less on services like education and health in order to cut taxes a lot.

(Conservative, Labor, Liberal Democrat) Party's View: Government should definitely not introduce a minimum wage because too many low paid workers would then lose their jobs.

(Conservative, Labor, Liberal Democrat) Party's View: Britain should do much more to keep its distance from the European Union.

D.4 AU (Monarchy)

Issue Related Items

- Which one of these best describes the current role of the Queen in relation to the Australian Governor General? - She chooses the Governor General, She appoints the Governor General only on the advice of the Prime Minister, She appoints the Governor General only on the recommendation of parliament, She has no role in choosing or appointing the Governor General
- Which one of these is true of the current role of the Governor-General in Australia – He performs only ceremonial duties, He can decide whether or not to dismiss the government, He acts only on the Queen's instructions, He controls the government (He performs only ceremonial duties)
- If the referendum on the republic is passed, which one of these would the role of the President be more like? – The current Governor General, The American President, The British Prime Minister, the current Australian Prime Minister (The current Governor General)
- If the referendum on the republic is passed, which one of these best describes how the Australian Prime Minister in the Republic can remove the President? - At any time without reporting to parliament, At any time but must later obtain approval from the House of Representatives, Only after a fair trial in parliament, The Prime Minister could not remove president.

Under the proposed change to a Republic, which of these, as far as you know, would definitely change.

- The Australian Flag (will not change)
- The National Anthem (will not change)
- The word 'Royal' in the names of the Royal Australian Navy and the Royal Australian Air Force (will change)
- Australia's participation in the Commonwealth Games (will not change)
- On the whole would you say the Liberal Party, or the Labor Party, is more in favour of Australia becoming a Republic?

General Politics Items

- Which one of these best describes Aden Ridgeway - Leader of the federal opposition, Leader of the Australian Democrats, An aboriginal senator in parliament, A justice of the high court, Or is he none of these
- Which one of these best describes Jennie George - A Labor member of federal parliament, Secretary of the Teacher's Federation, President of the ACTU, Leader of the Australian Worker's Union, Or is she none of these

- And now about social and welfare issues. On the whole would you say the Liberal Party, or the Labor Party, is more concerned about social and welfare issues, they are equally concerned, or do you not have much of an impression about that?
- And now thinking about business. On the whole would you say the Liberal Party, or the Labor Party, is closer to business, they are equally close or do you not have much of an impression about that?

D.5 UK (Crime)

(All questions are true/false)

Issue Related Items

- British courts are allowed to sentence a murderer to death (false).
- Britain has largest prison population in Western Europe (true).
- Britain has more people serving life sentences than rest of European Community put together (false).
- Possible to be tried by jury in local magistrate court (false).

General Politics Items

- Number of MPs about 100 (false).
- Longest time allowed between general elections is 4 years (false).
- Britain's electoral system is based on proportional representation (false).

D.6 UK (Monarchy)

(All questions are true/false)

- Princess Anne next in line to throne. (false)
- PM, not the queen, is Britain's Head of State. (false)
- Queen is head of Commonwealth. (true)
- Queen's duty to decide election date. (false)
- No PM takes office without being asked by the Queen. (true)
- Queen is head of Church of England. (true)
- Queen is head of Church of Scotland. (false)
- Britain has unwritten constitution (true).

D.7 EU 2007

- Which of the following countries is an official candidate to join the EU? - Romania, Montenegro, Croatia, Morocco (Croatia)
- Are the members of the European Parliament – Directly elected by the citizens of their country, Elected for three year terms, Elected by the parliament of their country, Appointed by their national head of government (directly elected by the citizens of their country)
- Are new EU laws in the field of employment adopted by - The European Commission, in some cases with the European Parliament, The Council
- of Ministers, in some cases with the European Parliament, The European Parliament by itself, The European commission by itself (The Council of Ministers, in some cases with the European Parliament)

- Is the EU's role regarding unemployment benefits to - Finance the member states' unemployment benefit systems, Decide the level and length of unemployment benefits in the member states, Require that the member states merge their unemployment benefit systems by 2010, Guarantee that all EU citizens have access to unemployment benefits where they live (Guarantee that all EU citizens have access to unemployment benefits where they live)
- Roughly a third of the EU's budget is devoted to one of the following. Is it ... ? - Helping the EU's less prosperous regions, Subsidizing the EU's fishing industry, Financing diplomatic missions abroad, Maintaining the EU's administration and bureaucracy (Helping the EU's less prosperous regions).
- By 2050, is the percentage of the adult EU population that is 65 or older projected to be - About one quarter of what it is now, About half of what it is now, About the same as what it is now, About twice what it is now, About four times what it is now (About twice what it is now)
- Which of the following is true of the amount of foreign aid given by the EU and its member states, combined, versus the amount given by the US? - The EU and its member states give roughly four times as much, The EU and its member states give roughly twice as much, The EU and its member states and the US give about the same amount, The US gives roughly twice as much, The US gives roughly four times as much (The EU and its member states give twice as much)
- On average, what percentage of the total spending by the governments of the EU member states for foreign aid? Is it - About 1%, About 5%, About 9%, About 13%, About 17% (About 1%)
- Which of the following countries does NOT possess nuclear weapons? - Pakistan, India, North Korea, Japan (Japan)

On a scale from 0 to 10, where 0 means as far left as possible, 10 means as far right as possible, and 5 is exactly in the middle, where would you place

- Nicholas Sarkozy (Greater than 5)
- Gordon Brown (Less than 5)

D.8 New Haven

- Which of the following would you say is closest to the population of the greater New Haven region? (550,000);
- During the 1990s, would you say the rate of job growth in the New Haven region was more, about the same, or less than the rest of the United States? (more);
- Which of the following would you say is the major source of revenue for most town governments in the Greater New Haven region? (Property taxes);
- Would you say the population of the city of New Haven increased, stayed the same, or decreased during the 1990s? (decreased);
- Does Connecticut law allow communities to share property tax revenues? (yes);
- Would you say those communities with the most valuable property tend to have the lowest property tax rates, average property tax rates, or the highest property tax rates? (lowest rates);
- Which of the following categories would you say the Federal Aviation Authority would use to describe Tweed New Haven Airport? (non-hub);
- Would you say that maintaining Tweed New Haven Airport at its current level of service would require any significant investment?(yes)

D.9 NIC 2

- Which of the following is closest to what you believe is true about global warming? (Caused by human activities like driving cars and burning fuel);
- Some people think that global warming is not really a problem so there is no need to do anything about it. Others think it's a serious problem and we need to act now.
- Where would you place the Democratic Party? ("It's a serious problems and we need to act now");
- Where would you place the Republican Party? ("Global warming is not really a problem so there is no need to do anything about it");
- As far as you know, does President Bush support or oppose recent international agreements to control greenhouse gases, or haven't you heard anything about this? (Oppose);
- As far as you know, does President Bush want to increase foreign aid, decrease foreign aid, or keep it the same, OR haven't you heard anything about this? (Increase);
- Out of every \$100 in federal budget, about how many dollars would you say goes to military spending? (\$20);
- Out of every \$100 in the federal budget, about how many dollars would you say goes to foreign aid? (\$1);
- For every \$100 in goods & services produced by the US, how many dollars worth would you say are sold to customers abroad? (\$12);
- In those African countries with the highest rates of infection with the AIDS virus, roughly how many adults out of every 100 would you say have AIDS or the AIDS virus? (30 or more);
- True or false - "The US has veto on WTO decisions" (False);
- True or false - "The US has veto on the UNSC" (True)

D.10 BTP03

(same as NIC2 questions)

D.11 BTP04

- How did John Kerry vote in the Senate on the resolution authorizing President Bush to go to war with Iraq? (Voted for);
- Which of the following countries now harbors the most Al Qaeda and Taliban fighters? (Pakistan);
- True or false - "Iraq was directly involved in the attacks on the WTC and the Pentagon on 9-11 2001" (False);
- True or false - "Large quantities of weapons of mass destruction have been found in Iraq" (False);
- True or false - "On average, prescription drugs cost more in Canada than in the US" (False);
- Which of the following was true of George W. Bush during the Vietnam War? (served in the TX Air National Guard);
- Which of the following was true of John Kerry during the Vietnam War? (decorated officer serving in Vietnam);
- A major destination of white collar jobs that have gone to other countries is? (India);
- Which of the following is closest to the number of Americans killed in Iraq since the war began? (1000)

D.12 San Mateo

- According to the Association of Bay Area Governments, between 1999 and 2006, what was the gap each year between the number of new homes needed and the number actually produced in San Mateo County? (About 1000);
- As of Sept 2007, about how much was the median price of a single-family home in San Mateo County? (\$950,000);
- As of 2006, about what percentage of County households could afford a median-priced house in San Mateo County? (12%);
- As of 2007 what was the average rent for a 2-bedroom apartment in San Mateo County? (\$1700);
- According to the 2006 San Mateo County Housing Needs Study, by about how many housing units is the demand for housing in San Mateo County projected to grow between 2005 and 2025? (70,000);
- According to the same Housing Needs Study, among the new households that are expected to form in the County between 2005 and 2025, about what percentage, combined, would be households with low, very low and extremely low-incomes? (About half);
- According to the 2006 San Mateo County Housing Needs Study, roughly how many new homes (including houses, townhomes, condos, and apartments) will be built between 2005 and 2025 at the current rate of housing development? (25,000);
- What percentage of San Mateo County's land is agricultural use, watershed, open space, wetlands, or parks? (more than 75%)

D.13 BTP05

- Among 29 wealthy industrialized nations, does the US rank top 3, top 10, bottom 10, or bottom 3 in students' math skills? (Bottom 10);
- In the last few years, it has become harder, easier or about the same to obtain US work permits for foreigners. (Harder);
- In the last few years, "the gap between minority and white students in math and reading tests at the elementary school level" has increased, decreased, or stayed about the same. (Increased);
- Number of uninsured Americans. Answer choices: 1 million, 25 million, 45 million, 65 million or 85 million.(45mill);
- "Over the next fifty years, do you think the number of Americans over age 65 will...?" Answer choices: tripled, doubled, increased by half, or stayed about the same (Doubled);
- The percentage of uninsured low income families. Answer choices: a quarter, fifty percent, two-thirds, or ninety percent (Two Thirds)

D.14 Bulgaria

(All of the questions were True or False)

- If a policeman wants to detain a citizen for questioning at any time of the day or night, the citizen does not have the right to refuse()
- The Court has the right to detain suspects as long as needed to prove if they are guilty or not()
- The prosecution can detain a suspect until the case is resolved()
- The Chief Prosecutor is responsible and accountable to Parliament()

- The police do not have the right to use violence against a detainee even if proven that he/she has committed a crime()
- The police can detain you for 72 hours()
- A confession is not enough to find the defendant guilty()

D.15 Texas Utility Poll (CPL)

- What energy source do you think produces the greatest amount of CPL's electricity; coal, wind, natural gas, fuel oil, nuclear, or solar? (Natural Gas)
- Overall, what group of CPL customers consume the most kilowatt hours of electricity; residential, commercial, or industrial? (Industrial)
- Which customer group pays the highest electric rates; residential, commercial, or industrial? (Commercial)
- Which of the following resources used to produce electricity produces the most air emissions; nuclear, coal, or natural gas? (Coal)
- Roughly, how much profit do Texas electric utilities make on fuel; 0%, 10%, 20%, or 30%? (0%)
- Which agency sets electric rates in Texas? – PUC, Texas Utility Commission, Government, CPL, WT, SWEPCO, Other. (PUC)
- How much of your electric bill goes for fuel; less than 20%, 21 to 30%, 31 to 40%, 41 to 50%, or more than 50%? (21 – 30%)